

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November – December 2011

EE – 208 Micro Electronics & Integrated Circuit

Date: 02/12/2011, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1 What is a transistor? Draw the symbols of PNP and NPN transistor and explain its three sections Emitter, Base and Collector. 7
- Q-2 A. Explain the Common Emitter connection of a transistor with neat circuit diagram and derive an expression for Current amplification factor - β (in terms of α) and Output current (I_C). 6
- B. Fig-1 shows that a silicon transistor with $\beta=100$ is biased by base resistor method. Determine the operating point (V_{CE} and I_C). (Take $V_{BE}=1$ V) 4
- C. What is the meaning of Faithful Amplification of BJT? What are the conditions must be satisfied for that? Discuss in short. 4

OR

- Q-2 A. What do you mean by Pinch off Voltage and Shorted-gate drain current in JFET? Explain with neat diagram. 4
- B. What are basic types of FET? Explain the Constructional details of JFET. 5
- C. Define the following terms for JFET. 5
- (1) a.c. drain resistance (2) Transconductance (3) Amplification factor
- Q-3 A. Explain Voltage Divider Biasing method with neat diagram. Also write down its Advantages and Disadvantages. Why it is universal method for Transistor biasing? 7
- B. Draw the h parameter equivalent circuit for Transistor. Find out Input impedance, Current gain and Voltage gain in terms of h parameters for Transistor. 7

OR

- Q-3 A. Find the h parameter of given circuit shown in Fig-2 with neat diagram. 6
- B. Explain working principle of JFET with neat diagram. 5
- C. Draw the Schematic symbols of following. 3
- (1) n-channel JFET (2) p-channel JFET (3) n-channel MOSFET

SECTION – II

- Q-4** Draw and explain the block diagram of op-amp. What are the features of the 741 op-amp IC? 7
- Q-5** A. Explain Summing, Scaling and Averaging Amplifiers in Inverting Configuration with neat diagram. 7
- B. Define the following terms. 3
- (1) CMRR (2) PSRR (3) Input bias current
- C. What are the ideal characteristics of an ideal op-amp? 4

OR

- Q-5** A. Explain V-I converter using grounded load. 5
- B. Explain the differential amplifier using one op-amp with neat diagram. 5
- C. Explain fixed voltage regulator. 4
- Q-6** A. Write merits and demerits of Active filters over passive filters. 3
- B. Explain the square wave generator with neat circuit diagram waveforms. 5
- C. Draw and explain the circuit of first-order low-pass filter. Draw its frequency response curve. 6

OR

- Q-6** A. Explain the pin configuration of 555 timer IC. 5
- B. Explain the working of 555 timer IC as an astable multivibrator with required waveforms. 5
- C. Design a low-pass filter at a cut-off frequency of 1 kHz with a passband gain of 2. 4
(Take $C = 0.01\mu F$).

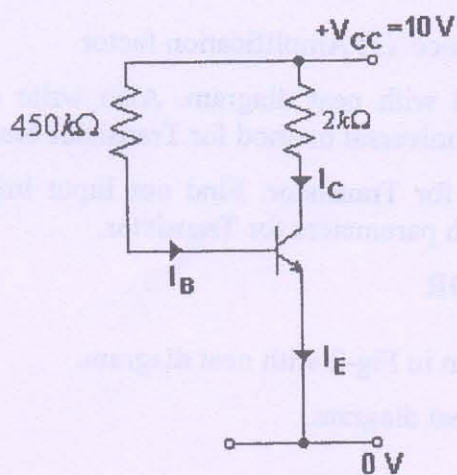


Fig-1

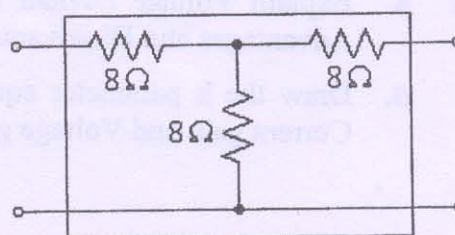


Fig-2